

Smart Garage Gate – Future Mall

Truth Table

Garage State	Car Count	Green LED	Red LED	Yellow LED	Blue LED
Empty	0	ON	OFF	OFF	OFF
Available	1–14	ON	OFF	OFF	OFF
Full	15	OFF	ON	OFF	OFF
Car Entered	0–14 → +1	ON / OFF according to status	ON / OFF according to status	BLINK	OFF
Car Exited	1–15 → -1	ON / OFF according to status	ON / OFF according to status	OFF	BLINK

System Rules

- The maximum number of cars is **15**.
- The minimum number of cars is **0**.
- The Green LED is ON when the garage has available spaces.
- The Red LED is ON only when the garage is Full.
- The Green and Red LEDs never turn ON at the same time.
- The Yellow LED blinks once when a car enters successfully.
- The Blue LED blinks once when a car exits successfully.